

TGIE — Financial Crime Evidence Package

Package EVP-TGIE-2026-0001-418aff52b0 · Case TGIE-2026-0001

SHA-256: 418aff52b031256b42a12636a23f1171bd103f3e7c915a13cb9a63e411f6c11c

Supporting Rules

```
[
  {
    "key": "amount",
    "label": "Transaction amount",
    "points": 25,
    "max": 25,
    "detail": "Largest transaction \u20b91,200,000"
  },
  {
    "key": "circular",
    "label": "Circular money flow",
    "points": 25,
    "max": 25,
    "detail": "Circular money flow (closed loop) detected"
  },
  {
    "key": "layering",
    "label": "Layering depth",
    "points": 22,
    "max": 22,
    "detail": "Forwarding chain depth 9"
  },
  {
    "key": "cash",
    "label": "Cash movement",
    "points": 18,
    "max": 18,
    "detail": "Cash movement combined with suspicious structure"
  },
  {
    "key": "payment_rails",
    "label": "Multiple payment rails",
    "points": 15,
    "max": 15,
    "detail": "6 payment rails used (CASH_IN, CASH_OUT, IMPS, NEFT, RTGS)"
  },
  {
    "key": "complexity",
    "label": "Network complexity",
    "points": 8,
    "max": 12,
    "detail": "15 accounts \u00b7 20 edges \u00b7 density 0.10"
  },
  {
    "key": "fan_in",
    "label": "Fan-in",
    "points": 7,
    "max": 20,
    "detail": "Fan-in from 4 sender account(s)"
  },
  {
    "key": "velocity",
    "label": "Transaction velocity",
    "points": 4,
    "max": 25,
    "detail": "3 transfer(s) within 120s"
  },
  {
    "key": "fan_out",
    "label": "Fan-out",
    "points": 4,
    "max": 20,
    "detail": "Fan-out to 3 recipient account(s)"
  }
]
```

MI Explanation

```
{
  "note": "ML ensemble explanation attaches when the model is scored for this case (Phase 5 wiring)."
}
```

Graph Metrics

```
{
  "accounts": 15,
  "edges": 20,
  "fan_out": 3,
  "fan_in": 4,
  "layering_depth": 9,
  "circular_transfers": true,
  "density": 0.19,
  "network_complexity": "High"
}
```

Path Analysis

```
{
  "longest_fund_path": [],
  "hops": 0,
  "edge_count": 20
}
```

Investigator Notes

```
{
  "notes": [],
  "comments": []
}
```

Regulatory Summary

```
{
  "suspicion_grounds": "Risk Score 100 (Critical). Contributing factors: Transaction amount (+25) \u00b7 Circular money flow (+25) \u00b7 Layering depth (+22) \u00b7 Cash movement (+18) \u00b7 Multiple payment rails (+15) \u00b7 Network complexity (+8).",
  "risk_score": 100,
  "confidence": 0.99,
  "accounts_involved": 15,
  "flagged_accounts": [],
  "transaction_count": 20,
  "total_amount": 8092000.0,
  "recommended_action": "File STR with FIU-IND"
}
```

Case Summary

```
{
  "case_id": "TGIE-2026-0001",
  "title": "High-Risk Transaction Cluster",
  "category": "High-Risk Network",
  "status": "Under Review",
  "priority": "Critical",
  "disposition": null,
  "assigned_to": "INV-2009",
  "assigned_name": "ARAV",
}
```

```
"department": null,
"created_at": 1782556524.435642,
"updated_at": 1782558534.526866
}
```

Timeline

```
[
{
  "id": "66126d5078",
  "ts": 1782558534.5268638,
  "event": "Case Assigned",
  "actor": "ARAV",
  "detail": "Assigned to ARAV"
},
{
  "id": "5302ebfc5b",
  "ts": 1782556525.435642,
  "event": "Case Created",
  "actor": "TGIE Risk Engine",
  "detail": "Risk 100/100 (Critical) \u00b7 15 accounts"
},
{
  "id": "34367a62e8",
  "ts": 1782556524.435642,
  "event": "Fraud Detected",
  "actor": "TGIE Risk Engine",
  "detail": "Risk Score 100 (Critical). Contributing factors: Transaction amount (+25) \u00b7 Circular money flow (+25) \u00b7 Layering depth (+22) \u00b7 Cash movement (+18) \u00b7 Multiple payment rails (+15) \u00b7 Network complexity (+8)."
}
]
```

Graph Snapshot

```
{
  "nodes": [
    {
      "id": "ACC_B2",
      "risk": 0,
      "role": "primary"
    },
    {
      "id": "ACC_RING3",
      "risk": 0,
      "role": "linked"
    },
    {
      "id": "ACC_B1",
      "risk": 0,
      "role": "linked"
    },
    {
      "id": "ACC_HUB",
      "risk": 0,
      "role": "linked"
    },
    {
      "id": "ACC_RING1",
      "risk": 0,
      "role": "linked"
    },
    {
      "id": "ACC_A1",
      "risk": 0,
      "role": "linked"
    },
    {
      "id": "CASH_SOURCE",
      "risk": 0,
      "role": "linked"
    }
  ]
}
```

```
"id": "ACC_RING2",
"risk": 0,
"role": "linked"
},
{
  "id": "ACC_ROOT",
  "risk": 0,
  "role": "linked"
},
{
  "id": "ACC_A2",
  "risk": 0,
  "role": "linked"
},
{
  "id": "CASH_EXIT",
  "risk": 0,
  "role": "linked"
},
{
  "id": "ACC_B3",
  "risk": 0,
  "role": "linked"
},
{
  "id": "ACC_B4",
  "risk": 0,
  "role": "linked"
},
{
  "id": "ACC_A3",
  "risk": 0,
  "role": "linked"
},
{
  "id": "ACC_FINAL",
  "risk": 0,
  "role": "linked"
}
],
"edges": [
  {
    "from": "CASH_SOURCE",
    "to": "ACC_ROOT"
  },
  {
    "from": "ACC_ROOT",
    "to": "ACC_A1"
  },
  {
    "from": "ACC_ROOT",
    "to": "ACC_A2"
  },
  {
    "from": "ACC_ROOT",
    "to": "ACC_A3"
  },
  {
    "from": "ACC_A1",
    "to": "ACC_B1"
  },
  {
    "from": "ACC_A1",
    "to": "ACC_B2"
  },
  {
    "from": "ACC_A2",
    "to": "ACC_B2"
  },
  {
    "from": "ACC_A2",
    "to": "ACC_B3"
  },
  {
    "from": "ACC_A3",
    "to": "ACC_B3"
  },
]
```

```

{
  "from": "ACC_A3",
  "to": "ACC_B4"
},
{
  "from": "ACC_B1",
  "to": "ACC_HUB"
},
{
  "from": "ACC_B2",
  "to": "ACC_HUB"
},
{
  "from": "ACC_B3",
  "to": "ACC_HUB"
},
{
  "from": "ACC_B4",
  "to": "ACC_HUB"
},
{
  "from": "ACC_HUB",
  "to": "ACC_RING1"
},
{
  "from": "ACC_HUB",
  "to": "ACC_FINAL"
},
{
  "from": "ACC_RING1",
  "to": "ACC_RING2"
},
{
  "from": "ACC_RING2",
  "to": "ACC_RING3"
},
{
  "from": "ACC_RING3",
  "to": "ACC_RING1"
},
{
  "from": "ACC_FINAL",
  "to": "CASH_EXIT"
},
},
"indicators": [],
"camera": null
}

```

Suspicious Accounts

```

[
  {
    "account": "ACC_B2",
    "role": null,
    "risk_score": null,
    "flagged": false
  },
  {
    "account": "ACC_RING3",
    "role": null,
    "risk_score": null,
    "flagged": false
  },
  {
    "account": "ACC_B1",
    "role": null,
    "risk_score": null,
    "flagged": false
  },
  {
    "account": "ACC_HUB",
    "role": null,
    "risk_score": null,
    "flagged": false
  }
]

```

```

},
{
  "account": "ACC_RING1",
  "role": null,
  "risk_score": null,
  "flagged": false
},
{
  "account": "ACC_A1",
  "role": null,
  "risk_score": null,
  "flagged": false
},
{
  "account": "CASH_SOURCE",
  "role": null,
  "risk_score": null,
  "flagged": false
},
{
  "account": "ACC_RING2",
  "role": null,
  "risk_score": null,
  "flagged": false
},
{
  "account": "ACC_ROOT",
  "role": null,
  "risk_score": null,
  "flagged": false
},
{
  "account": "ACC_A2",
  "role": null,
  "risk_score": null,
  "flagged": false
},
{
  "account": "CASH_EXIT",
  "role": null,
  "risk_score": null,
  "flagged": false
},
{
  "account": "ACC_B3",
  "role": null,
  "risk_score": null,
  "flagged": false
},
{
  "account": "ACC_B4",
  "role": null,
  "risk_score": null,
  "flagged": false
},
{
  "account": "ACC_A3",
  "role": null,
  "risk_score": null,
  "flagged": false
},
{
  "account": "ACC_FINAL",
  "role": null,
  "risk_score": null,
  "flagged": false
}
]

```

Transactions

```

[
  {
    "txn_id": "TXN-78F48C52B169",
    "amount": 1200000.0,

```

```

"date": 1782556524.435642,
"from_account": "CASH_SOURCE",
"to_account": "ACC_ROOT",
"rail": "CASH_IN",
"reason": "Transaction in flagged cluster"
},
{
"txn_id": "TXN-551A97F8EEC8",
"amount": 390000.0,
"date": 1782556524.435642,
"from_account": "ACC_ROOT",
"to_account": "ACC_A1",
"rail": "UPI",
"reason": "Transaction in flagged cluster"
},
{
"txn_id": "TXN-5C624C8D3866",
"amount": 395000.0,
"date": 1782556524.435642,
"from_account": "ACC_ROOT",
"to_account": "ACC_A2",
"rail": "IMPS",
"reason": "Transaction in flagged cluster"
},
{
"txn_id": "TXN-E64B0717C06A",
"amount": 400000.0,
"date": 1782556524.435642,
"from_account": "ACC_ROOT",
"to_account": "ACC_A3",
"rail": "NEFT",
"reason": "Transaction in flagged cluster"
},
{
"txn_id": "TXN-BA8994F5553F",
"amount": 190000.0,
"date": 1782556524.435642,
"from_account": "ACC_A1",
"to_account": "ACC_B1",
"rail": "UPI",
"reason": "Transaction in flagged cluster"
},
{
"txn_id": "TXN-04054C831019",
"amount": 195000.0,
"date": 1782556524.435642,
"from_account": "ACC_A1",
"to_account": "ACC_B2",
"rail": "RTGS",
"reason": "Transaction in flagged cluster"
},
{
"txn_id": "TXN-9577EE89D35B",
"amount": 195000.0,
"date": 1782556524.435642,
"from_account": "ACC_A2",
"to_account": "ACC_B2",
"rail": "UPI",
"reason": "Transaction in flagged cluster"
},
{
"txn_id": "TXN-A47F5B822E10",
"amount": 195000.0,
"date": 1782556524.435642,
"from_account": "ACC_A2",
"to_account": "ACC_B3",
"rail": "IMPS",
"reason": "Transaction in flagged cluster"
},
{
"txn_id": "TXN-3ABCA993939C",
"amount": 198000.0,
"date": 1782556524.435642,
"from_account": "ACC_A3",
"to_account": "ACC_B3",
"rail": "NEFT",
"reason": "Transaction in flagged cluster"
}

```

```

},
{
  "txn_id": "TXN-9737216E8395",
  "amount": 198000.0,
  "date": 1782556524.435642,
  "from_account": "ACC_A3",
  "to_account": "ACC_B4",
  "rail": "UPI",
  "reason": "Transaction in flagged cluster"
},
{
  "txn_id": "TXN-2A6A058F3547",
  "amount": 185000.0,
  "date": 1782556524.435642,
  "from_account": "ACC_B1",
  "to_account": "ACC_HUB",
  "rail": "UPI",
  "reason": "Transaction in flagged cluster"
},
{
  "txn_id": "TXN-1E51E0C207C5",
  "amount": 385000.0,
  "date": 1782556524.435642,
  "from_account": "ACC_B2",
  "to_account": "ACC_HUB",
  "rail": "IMPS",
  "reason": "Transaction in flagged cluster"
},
{
  "txn_id": "TXN-3BDEF2B95767",
  "amount": 388000.0,
  "date": 1782556524.435642,
  "from_account": "ACC_B3",
  "to_account": "ACC_HUB",
  "rail": "NEFT",
  "reason": "Transaction in flagged cluster"
},
{
  "txn_id": "TXN-E6B9348A0BAD",
  "amount": 193000.0,
  "date": 1782556524.435642,
  "from_account": "ACC_B4",
  "to_account": "ACC_HUB",
  "rail": "UPI",
  "reason": "Transaction in flagged cluster"
},
{
  "txn_id": "TXN-1EC062A6AFE4",
  "amount": 570000.0,
  "date": 1782556524.435642,
  "from_account": "ACC_HUB",
  "to_account": "ACC_RING1",
  "rail": "UPI",
  "reason": "Transaction in flagged cluster"
},
{
  "txn_id": "TXN-E8AC276A76C9",
  "amount": 570000.0,
  "date": 1782556524.435642,
  "from_account": "ACC_HUB",
  "to_account": "ACC_FINAL",
  "rail": "RTGS",
  "reason": "Transaction in flagged cluster"
},
{
  "txn_id": "TXN-A78CDF744AED",
  "amount": 565000.0,
  "date": 1782556524.435642,
  "from_account": "ACC_RING1",
  "to_account": "ACC_RING2",
  "rail": "IMPS",
  "reason": "Transaction in flagged cluster"
},
{
  "txn_id": "TXN-7BB7750EF376",
  "amount": 560000.0,
  "date": 1782556524.435642,

```

```

"from_account": "ACC_RING2",
"to_account": "ACC_RING3",
"rail": "NEFT",
"reason": "Transaction in flagged cluster"
},
{
"txn_id": "TXN-B943F7E88578",
"amount": 555000.0,
"date": 1782556524.435642,
"from_account": "ACC_RING3",
"to_account": "ACC_RING1",
"rail": "UPI",
"reason": "Transaction in flagged cluster"
},
{
"txn_id": "TXN-A03B61
... (truncated; see JSON package)

```

Risk Score

```

{
"risk_score": 100,
"assessment": {
"score": 100,
"level": "Critical",
"action": "Immediate investigation case \u00b7 recommend account freeze &
transaction hold",
"priority": "Critical",
"confidence": 99,
"factors": [
{
"key": "amount",
"label": "Transaction amount",
"points": 25,
"max": 25,
"detail": "Largest transaction \u20b91,200,000"
},
{
"key": "circular",
"label": "Circular money flow",
"points": 25,
"max": 25,
"detail": "Circular money flow (closed loop) detected"
},
{
"key": "layering",
"label": "Layering depth",
"points": 22,
"max": 22,
"detail": "Forwarding chain depth 9"
},
{
"key": "cash",
"label": "Cash movement",
"points": 18,
"max": 18,
"detail": "Cash movement combined with suspicious structure"
},
{
"key": "payment_rails",
"label": "Multiple payment rails",
"points": 15,
"max": 15,
"detail": "6 payment rails used (CASH_IN, CASH_OUT, IMPS, NEFT, RTGS)"
},
{
"key": "complexity",
"label": "Network complexity",
"points": 8,
"max": 12,
"detail": "15 accounts \u00b7 20 edges \u00b7 density 0.10"
},
{
"key": "fan_in",
"label": "Fan-in",

```

```

"points": 7,
"max": 20,
"detail": "Fan-in from 4 sender account(s)"
},
{
"key": "velocity",
"label": "Transaction velocity",
"points": 4,
"max": 25,
"detail": "3 transfer(s) within 120s"
},
{
"key": "fan_out",
"label": "Fan-out",
"points": 4,
"max": 20,
"detail": "Fan-out to 3 recipient account(s)"
},
],
"top_indicators": [
"Transaction amount",
"Circular money flow",
"Layering depth",
"Cash movement",
"Multiple payment rails"
],
"suppressed": false,
"suppression_reason": null,
"explanation": "Risk Score 100 (Critical). Contributing factors: Transaction amount (+25) \u00b7 Circular money flow (+25) \u00b7 Layering depth (+22) \u00b7 Cash movement (+18) \u00b7 Multiple payment rails (+15) \u00b7 Network complexity (+8).",
"should_create_case": true,
"investigation_threshold": 70,
"metrics": {
"accounts": 15,
"edges": 20,
"max_amount": 1200000.0,
"max_fan_out": 3,
"max_fan_in": 4,
"layering_depth": 9,
"circular": true,
"distinct_rails": 6,
"cash": true,
"density": 0.095
},
"scored_at": "2026-06-27T10:35:24.435284+00:00"
}
}

```

Fraud Pattern

```

{
"category": "High-Risk Network",
"detection_key": "det:ACC_A1|ACC_A2|ACC_A3|ACC_B1|ACC_B2|ACC_B3|ACC_B4|ACC_FINAL|ACC_HUB|ACC_RING1|ACC_RING2|ACC_RING3|ACC_ROOT|CASH_EXIT|CASH_SOURCE",
"fraud_dna": {
"dna_id": "FDNA-NET-086",
"type": "NET",
"signature": "0xD80B84",
"genes": [
{
"name": "Velocity",
"score": 26,
"label": "Low"
},
{
"name": "Amount",
"score": 70,
"label": "Rounded"
},
{
"name": "Structure",
"score": 95,
"label": "Dense cluster"
}
]
}

```

```

{
  "name": "Behavior",
  "score": 42,
  "label": "Established"
},
{
  "name": "Temporal",
  "score": 28,
  "label": "Business hours"
},
{
  "name": "Flow",
  "score": 41,
  "label": "Mixed flow"
},
{
  "name": "Risk",
  "score": 97,
  "label": "Critical"
},
{
  "name": "Outcome",
  "score": 50,
  "label": "Under investigation"
}
],
"behavior_summary": "Established",
"money_movement_style": "Circular / closed-loop laundering",
"layering_depth": 9,
"fan_out": 3,
"fan_in": 4,
"circular_transfers": true,
"dormant_accounts": 0,
"velocity": "Low",
"velocity_score": 26,
"network_complexity": "High",
"explanation": "This case carries a NET fraud-DNA signature. Its strongest behavioural genes are risk (97), structure (95), amount (70). The structure gene reads 'Dense cluster' and the money-flow gene reads 'Mixed flow', which is characteristic of this fraud methodology.",
"suggested_action": "Freeze Accounts Immediately",
"known_similar_cases": []
}
}

```

Reason

"Risk Score 100 (Critical). Contributing factors: Transaction amount (+25) \u00b7 Circular money flow (+25) \u00b7 Layering depth (+22) \u00b7 Cash movement (+18) \u00b7 Multiple payment rails (+15) \u00b7 Network complexity (+8)."

Confidence

0.99